

SUSHANT LOKHANDE

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EDUCATION

San Jose State University | *Master of Science in Computer Science* | GPA: 3.8/4.0 **Aug 2024 – May 2026**
Coursework: Distributed Systems, Database Systems, Advanced Algorithms, Cloud Computing, Machine Learning and AI

University of Mumbai | *Bachelor of Engineering in Computer Engineering* | GPA: 3.9/4.0 **Aug 2019 – May 2023**
Coursework: Data Structures & Algorithms, Object-Oriented Programming, OS, Computer Networks, Software Engineering

EXPERIENCE

San Jose State University | *Graduate Research Assistant* **San Jose, California | Aug 2025 – May 2026**

- Built a **scalable, production-grade Python and PyTorch pipeline** processing **50K+ image samples** across 3 model tracks, with profiling, batched GPU execution, and reproducible **Docker** builds for cloud-native deployment.
- Designed **distributed training and evaluation workflows** with **CloudWatch-style structured logging, monitoring**, automated regression checks, and **troubleshooting** runbooks, accelerating experiment iteration by **3×**.
- Wrote **clean, maintainable code** (pytest, type hints, **design patterns, object-oriented principles**), leveraged **GenAI and AI-powered tools (Cursor, Claude Code)** to enhance **developer productivity**, and co-authored a paper under review.

Turito | *Software Engineer Intern* **Hyderabad, India | Jun 2022 – Oct 2022**

- Shipped **12+ RESTful microservice endpoints** in **Python** on **AWS EC2** powering a **customer-facing edtech platform** serving **70K+ users globally**, with **90%+ pytest coverage** and **CI/CD** release pipelines.
- Architected **scalable, fault-tolerant batch and near-real-time data pipelines** ingesting **2M+ daily events** from **PostgreSQL (SQL)** into analytics views, cutting dashboard query latency from **8s to 1.2s**.
- Tuned **PostgreSQL** query plans, composite indexes, and **algorithm-level N+1** fixes on hot tables, improving **API p95 latency** by **55%** and lowering compute cost under live classroom load.
- Drove **code reviews, technical documentation, integration testing, and on-call duties** in a 4-person Agile team, resolving **20+ production incidents** and demonstrating **operational excellence** during peak traffic.

Trivia Softwares | *Software Engineer Intern* **Mumbai, India | Nov 2021 – Jan 2022**

- Built **Pandas/NumPy preprocessing pipelines** on **500K+ records** feeding an **ML inference service** across 15+ production dashboards, cutting downstream inference errors by **40%**.
- Owned **troubleshooting and resolving production issues** under **4-hour SLAs**, restoring backend and ML inference services with **resilient** recovery for end customers.

PROJECTS

Graph Connect – Multi-Datastore Microservice Platform | *Python, FastAPI, TypeScript, Neo4j, MongoDB, Docker, AWS*

- Designed a **scalable, cloud-native microservice** with **FastAPI, JWT auth**, and a **SQL/NoSQL multi-datastore (Neo4j + MongoDB)** modeling **50K+ relationships**, containerized with **Docker** for AWS deployment.
- Hardened with a **3-layer test suite** (Locust load, pytest regression, Selenium UI), **structured logging, and monitoring**, validating **fault-tolerant, resilient** behavior under simulated production load.

Melody Metrics – Distributed Data Pipeline | *C++, Python, Hadoop, MapReduce, Hive, Bash, Flask*

- Wrote **C++ MapReduce** jobs on a Linux **Hadoop** cluster ingesting **1M+ records** and fuzzy-joining heterogeneous sources in **Hive** into a unified 140K-row analytics dataset.
- Orchestrated the end-to-end **distributed pipeline** with Bash and exposed results via a Flask API, with downstream XGBoost analytics achieving **84% accuracy** on the joined dataset.

AI Learning Aggregator – RAG Knowledge Assistant | *Python, FastAPI, OpenAI, Vector Embeddings, Docker, AWS*

- Shipped a **GenAI-powered RAG pipeline** with ingestion, vector store, semantic retrieval, and **LLM grounding** for citation-backed responses, **Dockerized** and deployed on **AWS EC2**.
- Wrote **async Python (asyncio) FastAPI** services with prompt templates, context-window management, and guardrails on retrieval scope to control hallucination and improve factual accuracy.

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, Go, SQL, Bash, C

Cloud & Infrastructure: AWS (EC2, S3, Lambda, IAM), Docker, Kubernetes, Linux, GitHub Actions (CI/CD), microservices

Distributed Systems & Data: Hadoop, MapReduce, Hive, Spark, Kafka, ETL, distributed training

Databases (SQL & NoSQL): PostgreSQL, MySQL, MongoDB, Neo4j, Redis, Hive

Backend & APIs: FastAPI, Flask, Node.js, REST, microservices, JWT/OAuth, async I/O (asyncio), WebSockets, gRPC

AI/ML & GenAI: Scikit-learn, OpenAI, LLM integration, RAG, vector embeddings, prompt engineering, Cursor, Claude Code

Engineering & Quality: Data structures, algorithms, OOP, design patterns, SDLC, Agile/Scrum, code reviews, technical documentation, pytest, JUnit, Locust, monitoring, on-call

RESEARCH & HONORS

- Image-Based Techniques and Ensemble Soft Voting for Malware Classification** – *Under review*, Jun 2026.
- Spartan Scholar Scholarship** — San Jose State University, 2025. Awarded to top 3% of class for academic excellence.
- Comprehensive Big Data Analytics Dashboard on Chats** — Presented at **ICICN 2023**, Mumbai, India.